WORKED SOLUTION

STEP III 2012, Question 3

Pure • Curve sketching

STEP III 2012, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q3

The key idea

Substituting $y = 4 - x^2$ into the second equation gives a quartic in x whose roots are the x-coordinates of the meeting points. "Touching" means a *repeated* root. Instead of treating m and k as the unknowns, label everything by the x-coordinate a of the touching point: m, k and all the other meeting points then turn out to be simple functions of a. That tells us exactly what each of the four sketches must look like, and part (ii) is the special case $m = 12$.

Solution

Setting up: where can the curves touch?

Call the curves $C_1\colon y = 4 - x^2$ and $C_2\colon mx = k - y^2$. C_1 is the familiar downward parabola through $(\pm 2, 0)$ and $(0, 4)$. C_2 is $x = (k - y^2)/m$: a parabola symmetric about the x-axis, opening to the left (as $m > 0$), with vertex $(k/m, 0)$. It meets the y-axis at $(0, \pm\sqrt{k})$ when $k > 0$ and not at all when $k < 0$.

Every point of C_1 is $(x, 4 - x^2)$, so it lies on C_2 exactly when

$$F(x) = x^4 - 8x^2 + mx + 16 - k = 0,$$

since $(4 - x^2)^2 = x^4 - 8x^2 + 16$.

? **Ask yourself** Why does "touch" mean that F has a repeated root?

At a common point, C_1 has direction $(1, -2x)$ and C_2 (the level curve of $mx + y^2$) has normal $(m, 2y)$. The curves touch when the direction is perpendicular to the normal: $m - 4xy = 0$, that is $m - 4x(4 - x^2) = 0$. But $F'(x) = 4x^3 - 16x + m$ is exactly this expression. So the curves touch at $x = a$ precisely when $F(a) = F'(a) = 0$, a repeated root.

From $F'(a) = 0$ we get $m = 4a(4 - a^2)$. Substituting into $F(a) = 0$,

$$k = a^4 - 8a^2 + 4a^2(4 - a^2) + 16 = -3a^4 + 8a^2 + 16 = (4 - a^2)(3a^2 + 4),$$

and so

$$\frac{k}{m} = \frac{3a^2 + 4}{4a}.$$

Since $m > 0$ we need $0 < a < 2$ or $a < -2$. Dividing out the double factor,

$$F(x) = (x - a)^2(x^2 + 2ax + 3a^2 - 8),$$

as we can check by expanding: the x^3 terms cancel, the x^2 coefficient is $3a^2 - 8 - 4a^2 + a^2 = -8$, the x coefficient is $2a^3 - 2a(3a^2 - 8) = 4a(4 - a^2) = m$ and the constant is $a^2(3a^2 - 8) = 16 - k$. The quadratic factor has discriminant $8(4 - a^2)$, so there are other meeting points only when $0 < a < 2$. They are at

$$x_\pm = -a \pm \sqrt{8 - 2a^2}.$$

Now we can sort the four cases by a:

- $k < 0$ if and only if $a^2 > 4$, so $a < -2$.
- For $0 < a < 2$, $k - 16 = a^2(8 - 3a^2)$, so $k > 16$ if and only if $a < \sqrt{8/3}$.
- For $0 < a < 2$, $\frac{k}{m} < 2$ if and only if $3a^2 - 8a + 4 < 0$, i.e. $(3a - 2)(a - 2) < 0$, i.e. $a > \frac{2}{3}$.

So (a) is $a < -2$, (c) is $0 < a < \frac{2}{3}$, (d) is $\frac{2}{3} < a < \sqrt{8/3}$ and (b) is $\sqrt{8/3} < a < 2$. (No touching configuration has $0 < k < 16$ with $k/m > 2$, which is why that case isn't listed.)

Part (i)

In every case the axis intercepts are $(\pm 2, 0)$ and $(0, 4)$ for C_1, and $(k/m, 0)$, plus $(0, \pm\sqrt{k})$ when $k > 0$, for C_2.

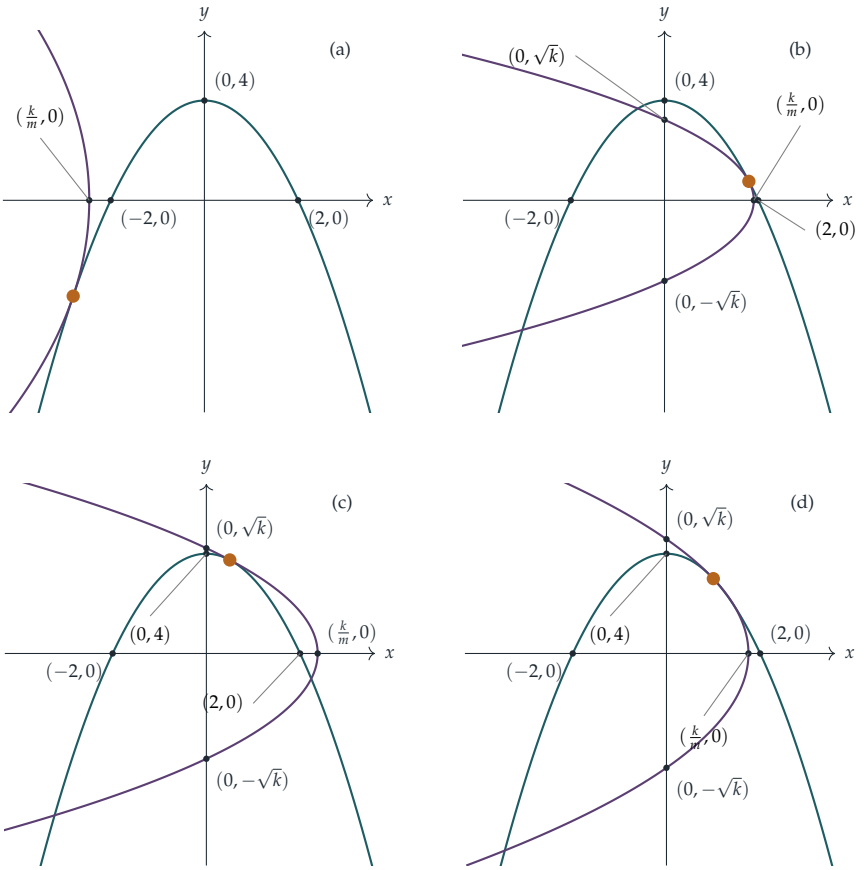
(a) Here $a < -2$, so the curves touch at $(a, 4 - a^2)$ in the third quadrant, and they don't meet anywhere else. The vertex is at $k/m = -(3a^2 + 4)/(4|a|)$, which is less than -2 because $(3|a| - 2)(|a| - 2) > 0$. So C_2 lies entirely to the left of $(-2, 0)$, and its lower arm rests on the left branch of C_1.

(b) Now $\sqrt{8/3} < a < 2$, so the touch is in the first quadrant, close to $(2, 0)$. The vertex has $\sqrt{3} \leq k/m < 2$ (by AM–GM, $3a^2 + 4 \geq 4\sqrt{3}\,a$), and $\sqrt{k} < 4$, so the vertex and both y-intercepts of C_2 lie under C_1. The upper arm touches C_1 from below and then crosses it at x_+, where $-2 < x_+ < 0$ (a point in the second quadrant). The lower arm crosses C_1 at $x_- < -2$, in the third quadrant.

(c) Here $0 < a < \frac{2}{3}$. The vertex is to the right of $(2, 0)$ and $\sqrt{k} > 4$, so C_2 wraps round the top of C_1 and touches it from outside in the first quadrant. Then $x_+ > 2$, because $\sqrt{8 - 2a^2} > 2 + a \iff (3a - 2)(a + 2) < 0$, and $x_- < -2$. So the lower arm crosses C_1 in the fourth quadrant and again in the third.

(d) Here $\frac{2}{3} < a < \sqrt{8/3}$. The vertex is between 0 and 2, under C_1, but $\sqrt{k} > 4$, so the upper arm passes above $(0, 4)$. It touches C_1 in the first quadrant and also crosses it there, at $x_+ \in (0, 2)$ (since $x_+ > 0 \iff a^2 < \frac{8}{3}$). The lower arm crosses at $x_- < -2$.

In every case $x_- < -2$ for $0 < a < 2$, because $a + \sqrt{8 - 2a^2} > 2$ (square $\sqrt{8 - 2a^2} > 2 - a$). The sketches use $a = -2.8,\ 1.8,\ 0.5,\ 1$ respectively. C_1 is teal, C_2 is plum and the touching point is amber.



Part (ii)

With $m = 12$, substituting $y = 4 - x^2$ into $12x = k - y^2$ gives $12x = k - 16 + 8x^2 - x^4$, that is

$$x^4 - 8x^2 + 12x + 16 - k = 0,$$

as required.

At the touching point $x = a$ this quartic has a repeated root (see the box above), so its derivative also vanishes there:

$$4a^3 - 16a + 12 = 0 \quad \Longrightarrow \quad a^3 - 4a + 3 = 0.$$

Since $a = 1$ is a root, $a^3 - 4a + 3 = (a-1)(a^2 + a - 3)$, and so

$$a = 1 \qquad \text{or} \qquad a = \frac{-1 \pm \sqrt{13}}{2}.$$

To find k, put $x = a$ in the quartic and reduce using $a^3 = 4a - 3$, so that $a^4 = 4a^2 - 3a$:

$$k = a^4 - 8a^2 + 12a + 16 = (4a^2 - 3a) - 8a^2 + 12a + 16 = -4a^2 + 9a + 16.$$

Now the three cases:

- $a = 1$: $k = 21$, so $k > 16$ and $k/m = \frac{7}{4} < 2$. This is case **(d)**.
- $a = \frac{1}{2}(-1 + \sqrt{13}) \approx 1.303$: here $a^2 = 3 - a$, so $k = -4(3-a) + 9a + 16 = 13a + 4 = \frac{1}{2}(13\sqrt{13} - 5) \approx 20.94$. This exceeds 16 because $13\sqrt{13} > 37$ (square: $2197 > 1369$), and it is below 24 because $13\sqrt{13} < 53$ ($2197 < 2809$). So $k > 16$ and $k/m < 2$, which is case **(d)** again.
- $a = \frac{1}{2}(-1 - \sqrt{13}) \approx -2.303$: $k = 13a + 4 = -\frac{1}{2}(5 + 13\sqrt{13}) \approx -25.94 < 0$. This is case **(a)**.

So only sketches (a) and (d) arise. This agrees with the classification by a above: 1 and 1.303 lie in $(\frac{2}{3}, \sqrt{8/3}) \approx (0.667, 1.633)$, and $-2.303 < -2$. Note that $a^3 - 4a + 3 = 0$ is just $m = 4a(4 - a^2)$ with $m = 12$.

The two (d) pictures differ slightly. For $a = 1$ the other meeting points are $x = -1 \pm \sqrt{6}$; the crossing at $x \approx 1.45$ is to the *right* of the touch, and the upper arm touches C_1 from above (as in the sketch). For $a \approx 1.303$ the crossing is at $x \approx 0.84$, to the left of the touch, and the upper arm touches from below.

Checks

- With $m = 12$, $k = 21$ the quartic factorises as $(x-1)^2(x^2 + 2x - 5)$. Expanding gives $x^4 - 8x^2 + 12x - 5$, which is the quartic with $16 - k = -5$.
- At $(1, 3)$ the gradient of C_1 is -2, and implicit differentiation of $12x = 21 - y^2$ gives $\mathrm{d}y/\mathrm{d}x = -6/y = -2$. So the curves really do touch there.
- Numerically, for random a in $(0, 2)$ and $a < -2$, the formulas $m = 4a(4 - a^2)$ and $k = (4 - a^2)(3a^2 + 4)$ make $F(a)$ and $F'(a)$ vanish, and the case boundaries fall at $a = \frac{2}{3}$ and $a = \sqrt{8/3}$. For $a \approx 1.3028$ a root search finds the other meetings at $x \approx 0.8433$ and $x \approx -3.4488$, as $x_\pm$ predicts.

Common slips

- Requiring only one root of the quartic. "Touch exactly once" allows other crossing points: in case (d) with $m = 12$ there are two.
- Drawing C_2 opening to the right. With $m > 0$, $x = (k - y^2)/m$ decreases as $|y|$ grows.

- In (a), forgetting that C_2 has no y-intercepts, or placing its vertex between -2 and 0. It must be to the left of -2.

Beyond STEP

Labelling the configurations by the touching point a is a first glimpse of *discriminants* and *envelopes*. The set of (m, k) for which F has a repeated root is the curve $\big(4a(4-a^2),\ (4-a^2)(3a^2+4)\big)$, and crossing it changes the number of real meeting points by two. This is the beginning of the algebraic geometry of plane curves, and of the singularity ("catastrophe") theory used to study how solutions change with parameters. Siklos's *Advanced Problems in Mathematics* has many more STEP-style curve problems in the same spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.